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Research paper

Drivers of habitat partitioning among three *Quercus* species along a hydrologic gradient

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A critical process that allows multiple, similar species to coexist in an ecological community is their ability to partition local habitat gradients. The mechanisms that underlie this separation at local scales may include niche differences associated with their biogeographic history, differences in ecological function associated with the degree of shared ancestry and trait-based performance differences, which may be related to spatial or temporal variation in habitat. In this study we measured traits related to water-use, growth and stress tolerance in mature trees and seedlings of three oak species (*Quercus alba* L., *Quercus falcata* Michx. and *Quercus palustris* Münchh.) which co-occur in temperate forests across the eastern USA but tend to be found in contrasting hydrologic environments. The three species showed significant differences in their local distributions along a hydrologic gradient. We tested three possible mechanisms that influence their contrasting local environmental distributions and promote their long-term co-existence: (i) differences in their climatic distributions across a broad geographic range, (ii) differences in functional traits related to water use, drought tolerance and growth and (iii) contrasting responses to temporal variation in water availability. We identified key differences between the species in both their range-wide climatic distributions (especially aridity index and mean annual temperature) and physiological traits in mature trees and seedlings, including daily water loss, hydraulic conductance, stress responses, growth rate and biomass allocation. Taken together, these differences explain the habitat partitioning that allows three closely related species to co-occur locally.

Keywords: coexistence, drought stress, niche partitioning, sapflow.

Introduction

Understanding the mechanisms that permit the coexistence of multiple closely related species in a community has been fundamental to ecological research for over a century (Connell 1961, May & MacArthur 1972, Volterra 1926). Local-scale abiotic heterogeneity in factors like water, nutrients, light or physical space affects the biotic community composition because performance differences among species provide competitive advantages in different parts of the gradient (Hutchinson 1959,

Silvertown 2004, Silvertown et al. 1999). Variation in a small number of abiotic factors can allow a large number of closely related species to coexist within a landscape (Cavender-Bares et al. 2004a). Temporal variation or intermittent stressors can provide an additional ‘axis’ on which species can partition local environments (Levins 1966, 1969, May and MacArthur 1972, Tilman 1994). The broad question of coexistence thus narrows: how do similar species exploit temporal and spatial variability to coexist sympatrically?

Trait-based approaches (e.g. Shipley et al. 2017) have focused on the ways that physiological differences between species affect community assembly processes or predict habitat preferences. Wei et al. (2017) identified several key traits in Salicaceae species that showed variation along a hydraulic gradient associated with fitness that predicted species distribution. The traits that support local diversity may operate at different timescales: plants can adjust solute concentration or stomatal opening within minutes and alter wood vascular growth or phenology along growing seasons or lifetimes (Chenu et al. 2008, Munns 2002). The magnitude of these responses may be very sensitive to changes in water availability, or may be coordinated to allow plants to maintain a relatively constant water status despite environmental changes (Meinzer et al. 2016), and different combinations of traits may affect similar fitness responses in an environment (Reich et al. 2003).

Biogeographic history may offer a macro-scale explanation for how species sort into local niches; species whose lineages stem from different climate regimes may sort into different local habitats where their ranges overlap (Ackerly 2003, Cavender-Bares et al. 2016). For example, Sedio et al. (2015) found that a plant's microhabitat on Barro Colorado Island, Panama, is associated with the climate of the region where it originated. This pattern is also seen in communities that sort along elevation gradients, reflecting the water availability and temperatures of climates of origin (Harrison et al. 2010).

Close relatives, which may be ecologically similar due to shared ancestry (Webb et al. 2002, Wiens and Graham 2005), are often expected to exhibit functional differences in resource use and/or stress tolerance that promote niche differentiation (Donoghue 2008). Research has shown that niche differentiation can occur directly through competition to meet similar resource requirements, or indirectly via density-dependent mortality (Gilbert and Webb 2007, Parker et al. 2015, Violle et al. 2011). Experimental tests in plants (e.g., Cahill et al. 2008) are equivocal, however, and there are many instances of plant communities without this pattern (e.g., Kluge and Kessler 2011), especially in cases where changing environments or stressors can drive similar species to cluster (Burns and Strauss 2011, Mayfield and Levine 2010).

In this study, we examined whether the local distribution patterns of three oak species showed evidence for differences in physiological and growth responses to gradients of water availability and stress that promote habitat (and thus niche) partitioning. We further considered these factors in relation to their biogeographic history and phylogenetic relatedness. *Quercus alba* L., *Quercus falcata* Michx. and *Quercus palustris* Münchh. are three forest canopy species at the Smithsonian Ecological Research Center near coastal Maryland. Red (*Q. falcata* and *Q. palustris*, *Quercus* section *Lobatae*) and white (*Q. alba*, *Quercus* section *Quercus*) oaks have been shown to have a long history of parallel and sympatric diversification in eastern North America

(Hipp et al. 2018), and the two lineages therefore coexist across the continent (Cavender-Bares et al. 2018). While these three species coexist broadly, they have been qualitatively described as being found in contrasting hydrologic niches (Gleason and Cronquist 1991, le Hardy de Beaulieu and Lamant 2006). We hypothesized that the apparent habitat partitioning along a hydrologic gradient shown among these three species could be demonstrated quantitatively. If large-scale patterns predicted local-scale partitioning, we hypothesized that partitioning can be explained by some combination of the following factors (see Figure S1 available as Supplementary Data at *Tree Physiology* Online),

- (i) Climatic envelopes of the full ranges of each species, especially considering moisture, are predictive of the distribution of species across the local elevation gradient in our study site.
- (ii) Species more able to tolerate water stress induced by temporal variation (i.e., a drought versus wet year) will occupy a more varied environmental range than less tolerant species.
- (iii) Trade-offs in growth and physiological (e.g., transpiration and water use) performance will emerge among species across the gradient, consistent with an interpretation of contrasting adaptive advantages at either end of the elevation gradient.

We further expected to see more habitat separation between *Q. falcata* and *Q. palustris* than between either of those species and *Q. alba*, as both are red oaks and thus more closely related to each other than either is to *Q. alba* (Cavender-Bares et al. 2004b), though with only three species we could not meaningfully quantify phylogenetic niche partitioning in this study system.

We focused on a set of traits, centered around sap flow measurements in mature trees and gas exchange measurements in seedlings, to understand the trait differences that might explain differences in distribution driven by water availability. In mature trees heat-dissipation measurement of xylem sap flow offers an approximation of whole-tree transpiration (Catovsky et al. 2002, Cohen et al. 1981, Granier 1987, Hogg et al. 1997, Ladefoged 1960); although concerns exist that it may fail to account for variable tree anatomy (Burgess et al. 2001, Clearwater et al. 1999, Delzon et al. 2004, Lu et al. 2000), this method continues to provide one of the best approaches for capturing water fluxes in mature trees (Poyatos et al. 2016). Overall, the selected traits indicate water use and hydraulic performance, growth and productivity and stress response.

Associating the variability in particular traits—even small differences—with both the local and broader ranges and phylogenetic relationships of these three species can help us to subsequently understand how spatially and temporally varied habitats allow close relatives to coexist.

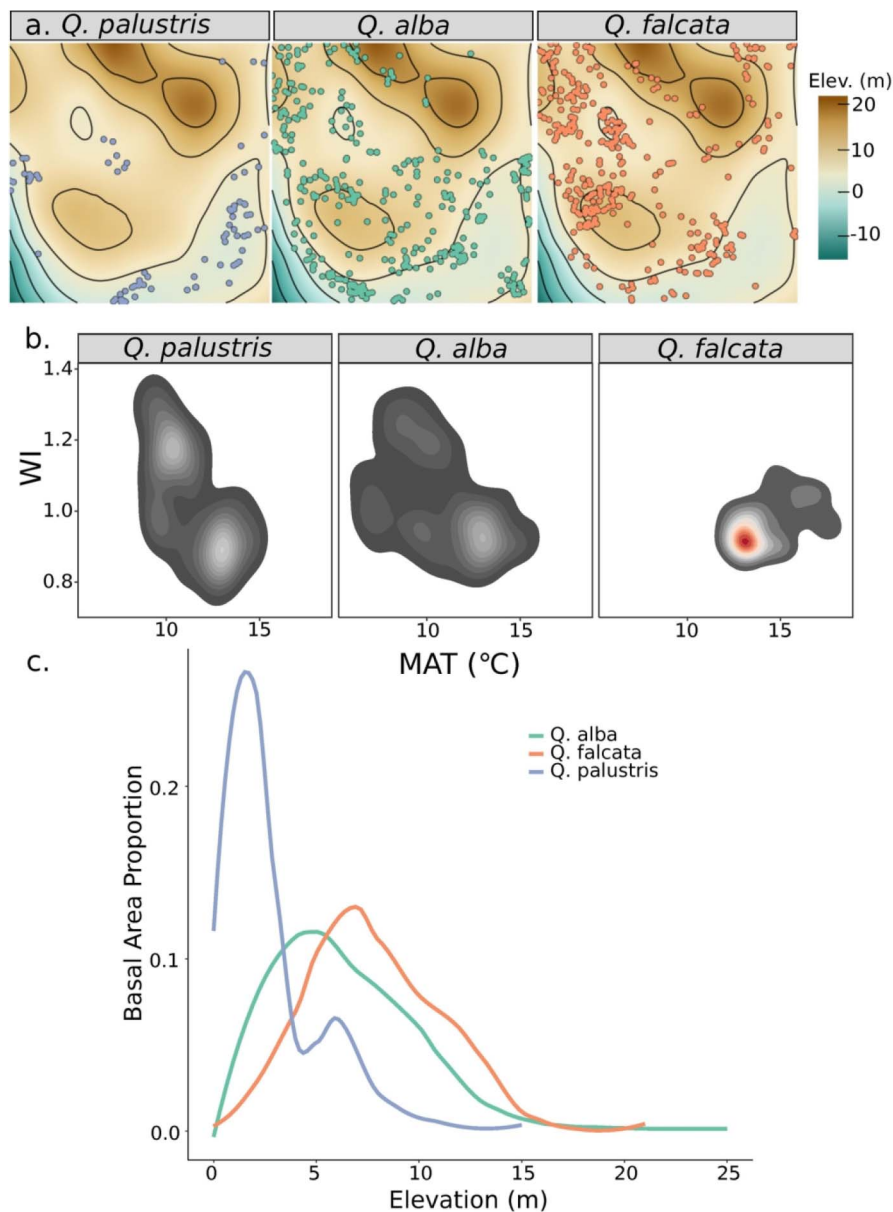


Figure 1. Broad and local distributions of *Q. alba*, *Q. falcata* and *Q. palustris*. (a) Distribution of the three oak species across SERC's BTP, 700 m × 700 m, with elevation (m above sea level). (b) The climatic envelopes of the North American ranges of the three study species. Wetness index, on the vertical axis, was developed by Zomer et al. (2008) and calculated as MAP over mean annual evapotranspiration. The horizontal axis is MAT in degrees Celsius. The distribution of *Q. falcata* is significantly drier (lower WI, $P < 0.001$) and hotter ($P < 0.001$) than the other two species. (c) Proportion of total basal area for each oak species found across the elevation gradient in the BTP. These distributions are further summarized in Table S3 available as Supplementary Data at Tree Physiology Online.

Materials and methods

Study system

Site description Our study sites for mature trees were located in a continuous tract of forest within the Smithsonian Environmental Research Center (SERC), located in Edgewater, Maryland, along the Rhode River. Our study site was an ~50 ha tract (the 'Big Tree Plot', BTP) within the 1100 ha main forest situated around the SERC photobiology tower (38.89° N, 76.56° W) and included an elevation gradient

ranging from sea level to 22 m, shown in Figure 1a. Mean slope angle at sampled sites was 6.69° (SD = 4.70°), and compound topographical index (CTI, Moore et al. 1991) was 3.62 (SD = 1.29). Historically used for dairy farming, the study area was reclaimed as forest ~100 years ago, and is currently made up of 50- to 100-year old stands in the 'tulip poplar' association. Since the time of data collection, a portion of this site has been added to the Forest Global Earth Observatory Network of the Center for Tropical Forest Studies (CTFS-ForestGEO).

Table 1. Mean soil moisture (volumetric water content, V_w/V_s) as measured by TDR probes in the mature tree field sites and seedling common garden. Differences among sites, years and treatments are all significant. Standard error (SE) is reported for each value in parentheses.

Mature trees		
	Water	SWC (SE)
Dry Year	Wet	0.186 (0.016)
	Mid	0.246 (0.017)
	Dry	0.157 (0.014)
Wet Year	Wet	0.466 (0.003)
	Mid	0.469 (0.005)
	Dry	0.485 (0.005)
Seedlings		
	Water	SWC (SE)
	Dry	0.124 (0.012)
	Mid	0.263 (0.017)
	Wet	0.462 (0.019)

The elevation gradient in the plot was treated categorically, such that trees sampled at 0–5 m were considered at low elevation and labeled ‘wet’ sites, 10–22 m were upland or ‘dry’ and those in between were at ‘mid’ elevation. These elevation categories are characterized by different soil types with contrasting hydrologic qualities (Soil Survey Staff 2018); six wells were installed in the BTP from 1 to 10 m in 2018, and the water table depth and soil moisture measurements show a strong relationship with elevation (see Figures S3 and S4 available as Supplementary Data at *Tree Physiology* Online). The low elevation sites were dominated by ‘Widewater and Issue’ (WBA) soils, a poorly drained soil with high flood frequency and a typical summer water table depth of 35 cm; the mean water table depth among all soil types at low elevation was 108 cm. Measured water table depth from May through September 2018 in wells at 1 and 2 m elevation ranged from 12 to 105 cm, with a mean of 46 cm. The mid-elevation sites included several soil types, typically moderate-to-well drained but with lower flood frequency than the low elevation sites. Mean water table depth estimated from soil types was 158 cm, while measured depths ranged from 16 (immediately following heavy rain) to 330 cm. The high elevation sites from which trees were sampled were primarily Collington and Annapolis (CRD) soils, which are well-drained, sandy soils with a summer water table depth of >200 cm, which is the reporting threshold for United States Geological Survey (USGS) soils data. Soil moisture data for the study site during the experiment are given in Table 1, and key differences in soil characteristics are summarized in Table 2. A map of the study trees including elevation and soil type can be found in Figure S2 (available as Supplementary Data at *Tree Physiology* Online).

Mature tree sampling Physiological measurements were collected from June to October 2002 and 2003 on the three

Table 2. Mean values of selected soil characteristics at SERC by elevation category. Available water storage (AWS) is the quantity of water available to plants for all soil layers. Available water capacity (AWC) is the water available for use by plants given in centimeters of water per centimeter of soil. Water table depth is the average from May to September, matching the months of measurements on mature trees; the maximum water table depth measured is 2 m and soils with a deeper water table were assigned this maximum value. Saturated hydraulic conductivity (K_{sat}) represents the rate of water movement through soil pores in a fully saturated soil. Soil organic matter (SOM) is measured as percentage by weight. Data were provided by the Web Soil Survey using a soil map of the minimum rectangular bounding box covering all trees measured in the experiment. All differences in means among elevation categories are significant (ANOVA, $P < 0.0001$).

	Wet (<5 m)	Mid (5–10 m)	Dry (>10 m)
AWS (cm)	31.0	31.1	26.9
AWC (cm cm ⁻¹)	0.17	0.16	0.14
Water table depth (cm)	108.2	157.8	>200
K_{sat} (μm s ⁻¹)	15.7	16.3	13.4
% SOM	1.28	0.64	0.38

most common oak species in the SERC forest tract: *Q. alba* L. (white oak), *Q. falcata* Michx. (Southern red oak) and *Q. palustris* Munchh. (pin oak). *Quercus palustris* is commonly found in floodplains with limited drainage, and is considered water-loving; *Q. falcata* grows in drier areas, including slopes and ridges above the floodplain. *Quercus alba* preferentially grows in more moderately watered areas and does not typically tolerate habitats with very high or very low water availability. Figure 1c shows the proportion of total basal area for each species found across the elevation gradient.

The number of trees sampled per species and elevation category in each year is summarized in Table S2 (available as Supplementary Data at *Tree Physiology* Online). Trees were all mature, with diameter-at-breast height (DBH) between 50 and 70 cm in similarly aged forest stands and were selected to cover the elevation range of the site, with considerations made for proximity to power sources. A map of study trees and elevation categories is included as supplemental Figure S2 (available as Supplementary Data at *Tree Physiology* Online). Soil moisture at each tree was measured approximately weekly each summer using time domain reflectometry (TDR): steel probes were installed to a depth of 1 m, and 5 cm apart, and readings were collected with a metallic cable tester (Tektronix 1502C, Tektronix Inc., Beaverton, OR, USA).

Sapwood area (SA, cm²) was measured in all sample trees. Tree cores were taken in late summer each year at a height of 1.4 m, away from the sap flow sensors and abnormal wood formations. Sapwood depth is the distance between the outermost ring of xylem and the point of color change marking the beginning of inactive heartwood. This approach was cross-validated with injection of Safranin-O dye. These cores were also used to estimate growth rate as basal area increment (BAI, cm²/year⁻¹). Average values of sapwood depth and DBH are

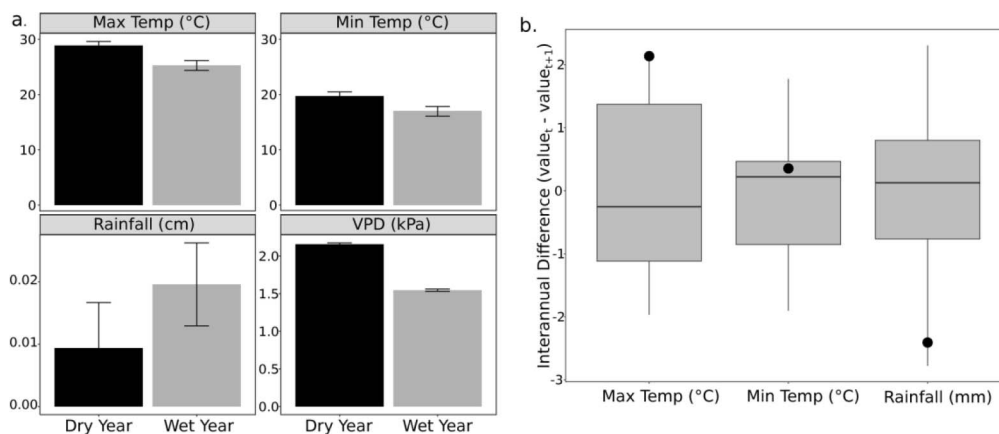


FIGURE 2. (a) Mean total daily rainfall, minimum and maximum daily temperatures and VPD for each summer during which data was collected, showing that dry summer was significantly warmer and drier than the wet summer. Error bars are $2 \times \text{SE}$. (b) Year-to-year difference in average summer minimum and maximum daily temperature and monthly rainfall totals, 1980–2010. The vertical axis gives the difference in average summer weather from year t and year $t + 1$. Dark bar is the mean, boxes are interquartile distance, and whiskers are 95% confidence interval. Dots highlight the difference in values between the study years, i.e., the value in 2003 minus the value in 2002.

reported in supplemental Table S4 (available as Supplementary Data at *Tree Physiology* Online).

Seedling common garden In addition to in situ measurements of mature trees, a common garden of oak seedlings was established. Acorns from each species were collected from within a 2 ha region of the SERC forest and planted in three blocks of an experimental garden. Seedlings were grown under 50% shade cloth for their first year of growth, then moved to an open-air rain-out shelter with an automatic irrigation system. Three water treatments were established: plants in the low water treatment were irrigated every 10 days, in the medium water treatment every 4 days, and in the high water treatment daily. Soil moisture was monitored using TDR probes; mean values are given for each treatment in Table 1.

Species geographic and climatic ranges The geographic ranges for each species were captured using occurrence data aggregated by the Global Biodiversity Information Facility (GBIF, 2016); only occurrences in USA with valid latitude and longitude coordinates were included. Climatic envelopes were produced for those ranges using the bioclimatic variables (chiefly mean annual temperature (MAT) and mean annual precipitation (MAP) generated by WorldClim Global Climate Data (Hijmans et al. 2005) and potential evapotranspiration (PET) and the aridity index (MAP over mean annual PET) data from the Consortium for Spatial Information (Trabucco and Zomer 2010, Zomer et al. 2008). We renamed aridity index as 'wetness index' (WI) for clarity. All data were processed using (R Development Core Team 2017).

Spatial and temporal variation in water availability

The SERC site falls within the humid subtropical climate zone, with warm summers and cool, wet winters. From 1990 to 2010, the

mean daily temperature in the summer was 21.9 (SD = 7.1), with mean monthly summer rainfall of 28.1 mm (SD = 11.2 mm, Global Historical Climatology Network Database), with a consistent pattern of interannual variation. Data were collected for this study over two summers with contrasting weather conditions. Weather during the 'dry year' (2002) saw significantly less rainfall, warmer temperatures, more solar radiation and higher vapor pressure deficit (VPD) than the 'wet year' (2003) (Figure 2a; for additional information see Table S1 available as Supplementary Data at *Tree Physiology* Online). The differences in these weather conditions from 2002 to 2003 were consistent with year-to-year differences for the region (Figure 2b). All site-specific climate data were collected at the climate monitoring station at SERC; measurements include global solar flux between 285 to 2800 nm (Eppley Precision Spectral Pyranometer, The Eppley Laboratory, Newport, RI, USA), temperature and relative humidity (Vaisala HMP45AC, Vaisala, Helsinki, Finland) and rainfall (TE525 'Tipping Bucket' rain gauge, Texas Electronics, Dallas, TX, USA). Saturated vapor pressure (VP_{sat}) and VPD were calculated from temperature and relative humidity as per the National Weather Service.

Trait measurements

The traits measured in mature trees and seedlings fall into three broad categories: water use, stress and growth/productivity. In mature trees, water use and stress traits were compared among species and across both space and time; growth rate was calculated as an average BAI over 20 years and was thus only compared among species and along the elevation gradient.

Water use: sap flow, conductance and water-use efficiency

We used constant-heat dissipation sap flow sensors consisting of a heated temperature sensor inserted into the sapwood

4 cm (per manufacturer recommendation) above an unheated reference temperature probe (Granier 1985, 1987). We used both commercial sensors (TDP 30; Dynamax, Inc., Houston, TX, USA) and constructed custom shorter sensors (Meinzer et al. 2004, Phillips et al. 2002); all probes used copper-constantin thermocouples to measure temperature. The current applied to constructed sensors was regulated by a circuit board to produce the same power density (wattage per unit volume of the resistor) to compensate for differences in resistance. The median temperature increase above ambient was 5.17°C for the Dynamax sensors and 8.24°C for the short sensors; this variation similar in magnitude to other studies (e.g., McCulloh et al. (2007)). Data from different sensor types showed no more variation than data from sensors of the same type in different positions in the tree (Figures S5 and S6 available as Supplementary Data at *Tree Physiology* Online). Sap flow was measured for 15 weeks beginning in mid-August in 2002 and 17 weeks beginning in early July in 2003.

Dynamax sensors were installed in each of 20 trees in both the dry and wet year; probes were inserted 0–30 mm into the cambium at 1.4 m in height on the north- and south-facing sides of each tree. In the wet year, when 19 additional trees were added to the study, two short sensors were also installed in each tree. These probes were inserted 11, 16 or 21 mm (depending on probe length) into the cambium; if Dynamax probes were already installed, the short probes were inserted at the same height 20 cm away. All sensors were insulated from water and heating, and connected to a current regulator (AVRD; Dynamax, Inc., Houston, Texas) and data logger (×23, ×21 or ×10; Campbell Scientific, Inc., Logan, UT, USA), powered by AC power with a battery backup. The temperature difference between the pairs of thermocouples for each probe were sampled every 10 s and averages were logged every 10 or 30 min. Data was downloaded weekly and potentially problematic data (due to malfunctioning sensors or electrical storms) was flagged.

Sap flow velocity (v , cm/s) was calculated from the maximum (ΔT_0) and actual (ΔT) temperature difference between probes at each time point, following the equations established by Granier (1987) and Granier et al. (1994) as shown in Eqs (3) and (4); volumetric sap flow rate (F , cm³/s⁻¹) is velocity multiplied by sapwood area. Total daily water loss (TDWL) was calculated as the integral of sap flow rate over a 24-h period and maximum sap velocity (V , cm/s⁻¹) is the maximum value of v in the same period.

$$v = 0.0119k^{1.231}, \quad (1)$$

where

$$k = \frac{\Delta T_0 - \Delta T}{\Delta T}. \quad (2)$$

Sap flow measurements taken with Dynamax sensors, which were generally longer than the sapwood depth, and thus in con-

tact with non-conducting tissue, were corrected per Clearwater et al. (1999) to account for overestimated sap flux velocity.

In seedlings, transpiration (E , mol H₂O m⁻² s⁻¹) was measured directly ($n = 108$, 12 plants per species per treatment) with a LI-COR 6400 Portable Photosynthesis System (LI-COR, Lincoln, NE, USA) rather than approximating via sap flux. Other water-use traits, including stomatal conductance (g_{sw} , mol H₂O m⁻² s⁻¹) and intrinsic water-use efficiency (WUE_i , A/ g_{sw}), were measured at the same time as E with the LI-COR 6400. Data were collected twice for each plant, over the last weeks of June and July, between 7:00 AM and 9:00 AM.

In both mature trees and seedlings, predawn (3:00 AM–6:00 AM) and midday (10:00 AM–3:00 PM) leaf water potential (Ψ_{PD} and Ψ_{MD} , respectively) were measured with a pressure chamber (Plant Water Status Console, 3000 series; Soilmoisture, Santa Barbara, CA, USA). Leaves for measuring Ψ_{PD} were taken from the most accessible location on tree, usually a low or mid canopy, while Ψ_{MD} was measured for high, mid and low canopy leaves. Between cutting and measurement, leaves were stored in moist, sealed plastic bags inside a dark cooler to minimize water loss. In mature trees, measurements were collected over the first three weeks of July in each year; in seedlings measurements were taken in the last two weeks of July.

In mature trees, the change in water potential from predawn to midday and steady-state sap flow rate (F) were used to calculate whole-plant hydraulic conductance (K , cm² s⁻¹ MPa⁻¹): $K = F/(\Psi_{MD} - \Psi_{PD})$. Steady-state F is the average F over the hour-long period during mid-day leaf collection when the variance in F was smallest. If sap flow measurements were not available for a tree on the date water potential was measured, the value from the most recent day with similar VPD was used. Whole-plant hydraulic conductance in seedlings (K_l , mmol m⁻² s⁻¹ MPa⁻¹) was similarly calculated, using steady state transpiration instead of sap flow rate.

The WUE (A/E) in seedlings was calculated directly from gas exchange measurements. In mature trees, it was estimated from $\delta^{13}C$ values (Farquhar et al., 1982), although this approach may be confounded by unknown mesophyll conductance (Warren & Adams, 2006). In each monitored mature tree, leaves were collected at multiple canopy positions at four time points across both summers. Leaf samples were frozen upon collection and then dried and ground for carbon isotopic analyses with an elemental analyzer (Carlo Erba Instruments, NA 2500 series, Wigan, UK) coupled via continuous flow to a stable isotope ratio mass spectrometer (ConFlo II to Delta Plus XL; ThermoFinnigan, Waltham, MA, USA) in the lab of Marilyn Fogel at the Geophysical Lab, Carnegie Institution of Washington, DC. Stable carbon isotopic values are expressed as δ values according to the equation:

$$\delta^{13}C = [(R_{sample}/R_{standard}) - 1]1000, \quad (3)$$

where R is the ratio of ^{13}C to ^{12}C and the standard was the Pee Dee Belemnite standard. Instrument error was $\pm 0.3\text{‰}$. When making comparisons among species and water availability, only mature mid-summer leaves for each tree were analyzed to minimize the problem of early, heterotrophic growth influencing WUE estimates (Terwilliger et al., 2001).

Water stress The change in daily leaf water potential ($\Delta\Psi$) between midday and predawn was used to estimate leaf water stress, where lower values suggest a leaf is closing its stomata or otherwise conserving water during the day and higher values show more water loss relative to the equilibrium indicated by the predawn water potential.

While measuring pre-dawn Ψ , maximum quantum yield of photosynthesis after dark incubation (F_v/F_m) was measured on seedlings with a portable chlorophyll fluorometer (MINI-PAM; Heinz Walz GmbH, Effeltrich, Germany). An indicator of the efficiency of photosynthesis, F_v/F_m is generally 0.8 in healthy plants and declines as plants experience stress (Maxwell & Johnson, 2000).

Growth and productivity The BAI ($\text{cm}^2/\text{year}^{-1}$) from tree cores collected in 2002 was used in mature trees to compare growth rates among species across the elevation gradient. Basal area was estimated from 1980 to 2002 by subtracting all newer ring growth from the present DBH; average growth rate was the slope of least-squares regression between year and basal area.

Productivity was directly measured in seedlings as carbon assimilation (A , $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$) with the LI-COR 6400 ($n = 108$, 12 plants per species per treatment, as with E). At the conclusion of the experiment, seedlings were harvested for biomass measurements. Leaves, stems and roots were separated and dried at 70°C for 3 days before weighing. In addition to biomass, leaf stable isotope ratios were measured following the same protocol as for mature trees.

Statistical analysis

All statistical analyses were performed in (R Development Core Team, 2017). For each physiological trait measured in mature trees, we tested the effects of and interactions among species, elevation category and year using analysis of variance (ANOVA) on a fixed effects model including all three attributes and all interactions. The same approach was used for the seedling trait data, with Species and Treatment as the explanatory variables. When ANOVA results were significant ($P < 0.05$), Tukey's Honest Significant Differences (HSD) was used to make pairwise comparisons.

Species climatic ranges were extracted from the raster BIOCLIM and aridity datasets at the coordinates of individuals identified in the GBIF data set using the *raster* package (Hijmans, 2016). Coordinate system corrections and conversion of shapefile data to raster format were done using QGIS

software (QGIS Development Team, 2009). Pairwise species means were compared using a Tukey-adjusted t-test for multiple comparisons (Lenth, 2016). All figures were produced using the *ggplot2* package (Wickham, 2009).

Results

Comparison of broad and local distributions

Comparing the climatic distributions of the oak species across their full ranges, the three species differed in mean bioclimatic variables in a few critical ways that were suggestive of associations with their local distributions at SERC. In particular, *Q. falcata* had a more arid climatic distribution and was found in the sites that have the lowest water availability locally. Across its range, *Q. falcata* was found in locations that were significantly warmer than either *Q. alba* or *Q. palustris*, which were generally not different from each other. The average MAT, in *Q. falcata*'s range was 3.2°C warmer than the range of *Q. alba* and 2.8°C warmer than *Q. palustris* ($P < 0.0001$, Tukey-adjusted P -value). *Quercus falcata* also occurred in regions with higher rainfall than the other species, with an average MAP 79 mm higher ($P < 0.0001$, Tukey-adjusted P -value), but because the higher temperatures drove a higher rate of PET, its range had a lower WI value, indicative of a drier environment overall. The temperature and aridity distributions of these three species are shown in Figure 1b. For climate variables, the differences between *Q. alba* and *Q. palustris* were not significant; these and additional bioclimatic variables are summarized in Table 3.

Figure 1c shows the distributions of each species of oaks for all trees in the BTP at SERC; these distributions are also summarized in Table S3 available as Supplementary Data at *Tree Physiology Online*. Though all three species were found across the elevation gradient (from the sea level floodplain to 22 m), each was concentrated in a distinct subset of the gradient from floodplain to higher elevation. Comparing the least-squares means of elevation by species, *Q. palustris* had the lowest mean elevation, showing a preference for locations where soils had

Table 3. Mean values of bioclimatic and soil hydrology variables in the North American ranges of *Q. alba*, *Q. falcata* and *Q. palustris*.

	<i>Q. alba</i>	<i>Q. falcata</i>	<i>Q. palustris</i>
Wetness (WI)	1.022	0.959	1.023
PET	1077.1	1219.6	1077.9
MAT (mm)	11.12	14.28	11.51
Min. T ($^\circ\text{C}$)	-7.08	-3.08	-6.71
Max. T ($^\circ\text{C}$)	29.39	31.48	29.87
MAP (mm)	1091.16	1168.00	1090.59
Wettest month (mm)	112.86	122.84	112.51
Driest month (mm)	68.71	74.49	67.14
Permeability (inches h^{-1})	3.43	3.53	3.57
Flood frequency	3.77	3.70	3.67

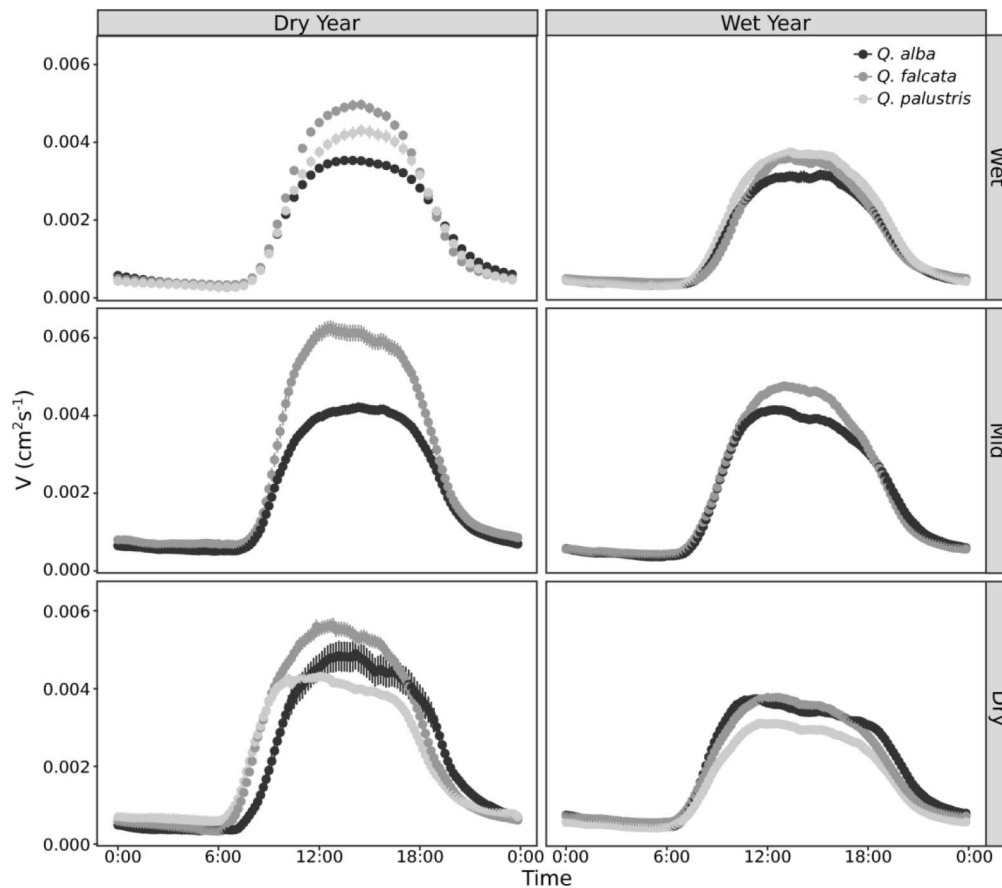


Figure 3. Average daily sap flow patterns for each species at each elevation and in the wet (July–September) and dry (August–October) year; the error bars are standard error. Daily maximum velocity is more variable in the dry year (SD in wet year = 0.00194, SD in dry year = 0.00239; $P = 0.0008$, Welch's two-sample t -test), and at low elevation (SD at low elevation = 0.00246, SD at high elevation = 0.00169; $P < 0.0001$, Welch's two-sample t -test).

higher water availability. As predicted, the largest difference in elevation was between the two red oaks, though both also occurred at significantly different mean elevations from *Q. alba* ($P < 0.0001$, Tukey-adjusted P -value).

Species, spatial and temporal performance differences

Water use Figure 3 shows the sap velocity over the course of 24 h averaged over the season (August–October in the dry year, July–September in the wet year) for each species; mean sap velocity for each species over time is shown in Figure S7 (available as Supplementary Data at *Tree Physiology* Online). The average maximum daily sap velocity (V , cm/s) was significantly different among species, elevation categories and years; each interaction between pairs of variables was also significant (ANOVA, $P = 0.0251$ for the species by year interaction, $P < 0.0001$ for all others). Mean values of V and other key traits measured in mature trees by species, elevation category, and year are summarized in Table 4.

Maximum sap velocity was also slightly but significantly correlated with both VPD (Pearson's $r = 0.128$, $P < 0.0001$)

and with solar radiation (Pearson's $r = 0.107$, $P < 0.0001$). The relationship between sap velocity and VPD was found in each species when considered separately and P -values were adjusted with the Holm method for multiple comparisons (*Q. alba*: $r = 0.120$, $P < 0.0001$; *Q. falcata*: $r = 0.159$, $P < 0.0001$; *Q. palustris*: $r = 0.105$, $P = 1.343 \times 10^{-4}$). The relationship between sap velocity and solar radiation persisted in both *Q. alba* and *Q. falcata* when these species were considered separately, but not in *Q. palustris* ($r = 0.0008$, $P = 0.97$).

Average TDWL is shown in Figure 5a. TDWL was significantly lower at low elevation than at either middle or high elevation sites ($P = 0.013$ and $P = 0.046$, respectively, Tukey's HSD); mean values did not differ among species or between years. There was a significant ($P = 0.024$) interaction between elevation and species; *Q. palustris* exhibited a decline in water loss with increased elevation (and therefore decreased soil moisture), while both *Q. alba* and *Q. falcata* generally exhibited increased water loss. In *Q. alba*, peak TDWL in both years occurred at high elevation sites (dry), while in *Q. falcata* peak values were seen at mid-elevation.

Table 4. Mean values of selected physiological traits measured in mature trees. Significance stars are the result of an ANOVA test and correspond to the following P -values: $p < 0.1$; $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Significant interactions between variables are described in Figure 5. V: maximum sap flux velocity; TDWL: total daily water loss; K: hydraulic conductance; $\Delta\Psi$: change in leaf water potential from pre-dawn to mid-day; $\delta^{13}\text{C}$: water-use efficiency.

Trait	Species			Site				Year	
	<i>Q. alba</i>	<i>Q. falcata</i>	<i>Q. palustris</i>	Wet	Mid	Dry		Dry	Wet
V (cm s^{-1})	4.29×10^{-3}	5.18×10^{-3}	4.51×10^{-3}	4.51×10^{-3}	5.09×10^{-3}	4.66×10^{-3}	**	4.91×10^{-3}	4.55×10^{-3}
TDWL (l)	13.17	12.28	12.28	8.72	15.56	14.07		12.83	12.47
K ($\text{cm}^2 \text{s}^{-1} \text{MPa}^{-1}$)	945.14	814.15	573.38	511.77	1165.68	744.52	**	876.96	729.65
$\Delta\Psi$ (MPa)	1.58	1.43	1.68	1.61	1.48	1.60	.	1.66	1.47
$\delta^{13}\text{C}$ (‰)	-29.190	-29.241	-28.645	* -29.002	-29.142	-28.840		-28.791	-29.138

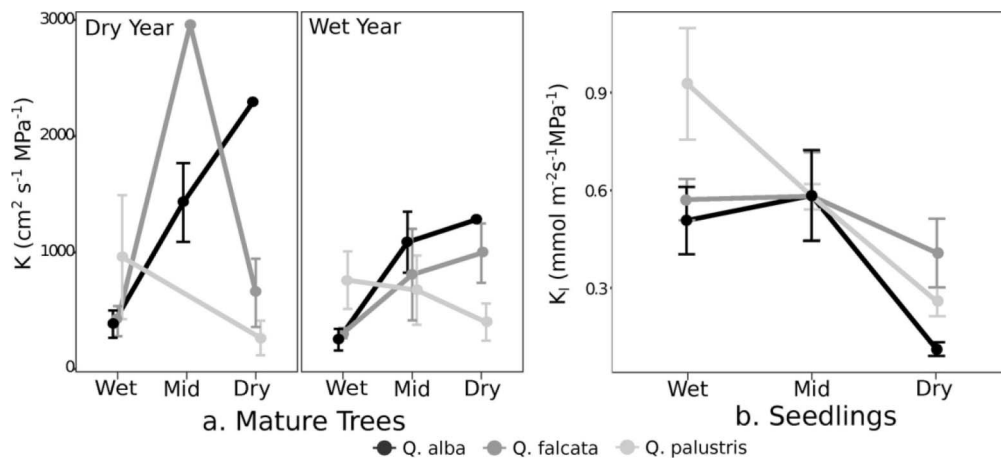


Figure 4. Differences in hydraulic conductance with water availability in mature trees (a) and seedlings (b). In mature trees, K ($\text{cm}^2 \text{s}^{-1} \text{MPa}^{-1}$) differed significantly by site-based water availability (ANOVA, $P = 0.00225$); interactions between water availability and species (ANOVA, $P = 0.003$) and the three-way interaction between year, elevation and species (ANOVA, $P = 0.03$) were also significant. In seedlings, conductance varied significantly by species (ANOVA, $P < 0.0001$ but not by water availability).

In mature trees, whole-plant hydraulic conductance (K) was similar to TDWL in its relationships with species, elevation and year (Figure 4a); there were significant differences in mean values among elevation categories ($P = 0.002$) but not species ($P = 0.38$) or year ($P = 0.13$). Much of this difference was driven by the much higher K seen in *Q. falcata* growing at mid-elevation in the dry year. There was also a significant interaction ($P = 0.003$) between species and elevation: among both *Q. alba* and *Q. falcata*, the lowest values of hydraulic conductance occurred in the wet floodplain, while in *Q. palustris*, trees in the floodplain had higher values of K than at other elevations.

Contrasting with mature trees, seedlings of all three species showed a significant decline in hydraulic conductance under the low water treatment; mean differences among species were not significant, and there was no significant interaction between species and water treatment when seedlings of all three species were included (Table 5). *Quercus palustris* does show a steeper decline in hydraulic conductance than *Q. falcata* (Figure 4b), but this difference is marginal ($P = 0.07$).

In mature trees, the strongest predictor of increased $\delta^{13}\text{C}$ ratios (and thus increased WUE) was leaf developmental stage indicated by calendar day. There were significant differences in $\delta^{13}\text{C}$ by species ($P = 0.028$) and marginal differences by

year ($P = 0.07$). There was also a change in the relationship between WUE and elevation between the years: on average, $\delta^{13}\text{C}$ increased when comparing wet to dry sites in the dry year and decreased in the wet year ($P = 0.023$). *Quercus palustris* had a higher WUE than the other two species in both years and at both wet and dry sites. It also exhibited the largest change in WUE at dry sites between the dry and wet years, as seen in Figure 5c.

Seedlings exhibited significant differences in transpiration rates (E) and stomatal conductance (g_{sw}) when grown under different water-availability conditions (E : $P < 0.0001$, g_{sw} : $P < 0.0001$), shown in Figure 7. Performance in all species was highest at the moderate water treatment for both traits; transpiration and g_{sw} were significantly higher at moderate water availability than either dry or wet conditions ($P < 0.0001$ for both traits, Tukey's HSD). There were no significant differences in either A or E among species, though g_{sw} was significantly higher in *Q. alba* than in the other two species ($P = 0.001$, Tukey's HSD).

Water-use efficiency (WUE) in seedlings was measured using both carbon isotope ratio ($\delta^{13}\text{C}$) and gas exchange measurements (A/g_{sw} , abbreviated as WUE_i). There were significant differences among species ($P = 0.04$) and treatments

Table 5. Seedling data means for a variety of physiological traits. Significance stars are the result of an ANOVA test and correspond to the following *P*-values: $p < 0.1$; $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Total biomass, leaf area, photosynthesis (*A*), transpiration (*E*), stomatal conductance (*g_{sw}*) and *F_v/F_m* each also show significant interactions between species and treatment.

Trait	Species				Treatment			
	<i>Q. alba</i>	<i>Q. falcata</i>	<i>Q. palustris</i>		Dry	Moderate	Wet	
Total biomass (g)	10.36	8.96	10.89	*	7.28	14.05	8.59	***
Proportion belowground	0.664	0.627	0.605	***	0.625	0.645	0.608	**
Leaf area (cm ²)	201.41	219.26	288.95	***	186.25	325.78	200.04	***
<i>A</i> (μmol CO ₂ m ⁻² s ⁻¹)	11.71	11.20	11.31		11.42	11.99	10.29	***
<i>E</i> (mol H ₂ O m ⁻² s ⁻¹)	3.64	3.58	3.51		3.46	3.91	3.20	***
<i>g_{sw}</i> (mol H ₂ O m ⁻² s ⁻¹)	0.177	0.159	0.162	**	0.149	0.190	0.153	***
WUE _i (A/ <i>g_{sw}</i>)	75.21	77.54	74.73		81.52	72.81	70.31	***
δ ¹³ C (‰)	-28.780	-29.666	-29.632	*	-28.752	-29.717	-29.688	*
<i>K_f</i> (mmol m ⁻² s ⁻¹ MPa ⁻¹)	0.422	0.519	0.593		0.272	0.582	0.724	***
ΔΨ	11.55	9.13	9.23	.	11.29	10.66	6.91	***
<i>F_v/F_m</i>	0.807	0.784	0.767	*	0.811	0.804	0.749	***

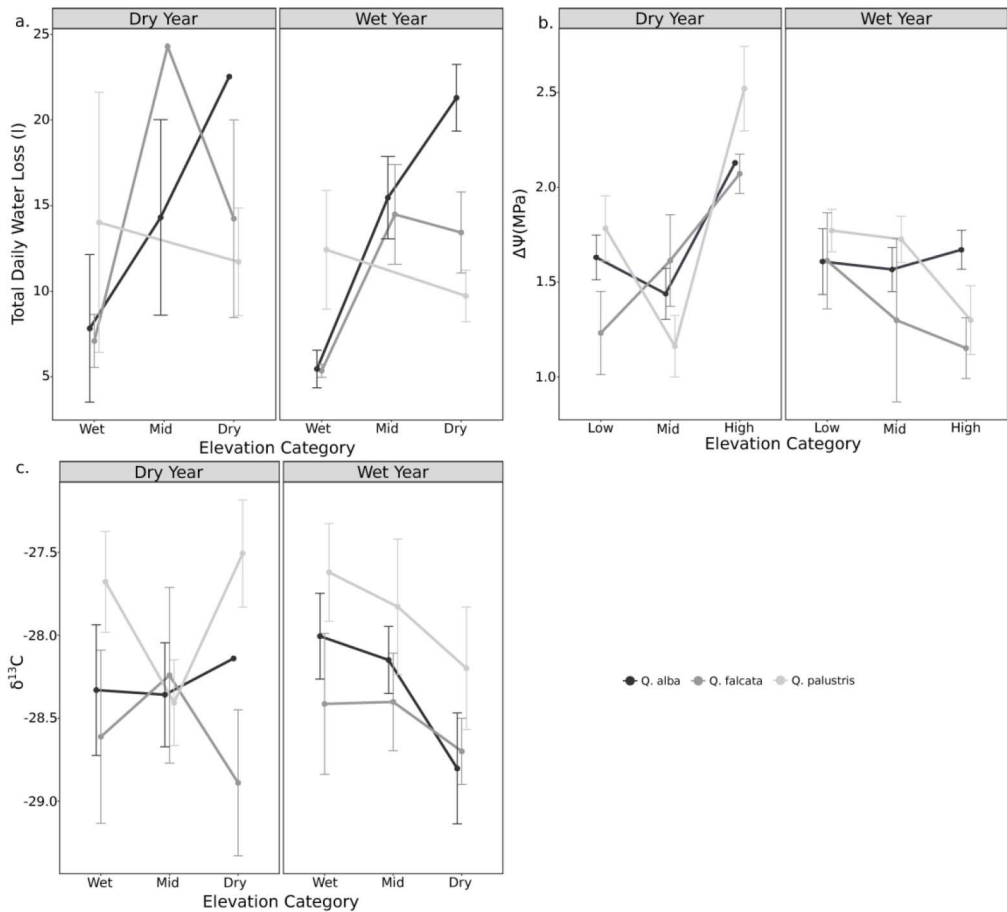


Figure 5. Differences by species, year and elevation in four water-use traits measured in mature trees. (a) There were significant differences in TWDL (l) by elevation (ANOVA, $P = 0.009$), as well as a significant interaction between elevation and species (ANOVA, $P = 0.024$). (b) ΔΨ was significantly higher in the drought year than in the wet year (ANOVA, $P = 0.037$), and there was a significant interaction between elevation category and year (ANOVA, $P < 0.0001$). (c) There were significant species differences (ANOVA, $P = 0.0167$) in WUE (δ¹³C) by species.

($P = 0.045$) in δ¹³C, though pairwise differences were small. *Quercus falcata* and *Q. palustris* were the most similar in their isotope ratios ($P = 0.99$), while *Q. alba* showed slightly, though non-significantly, higher WUE by this metric. WUE_i was not

significantly different among species but was among treatments, and was highest under dry conditions ($P < 0.0001$, Tukey's HSD). This pattern was also seen with δ¹³C, though the results were not significant.

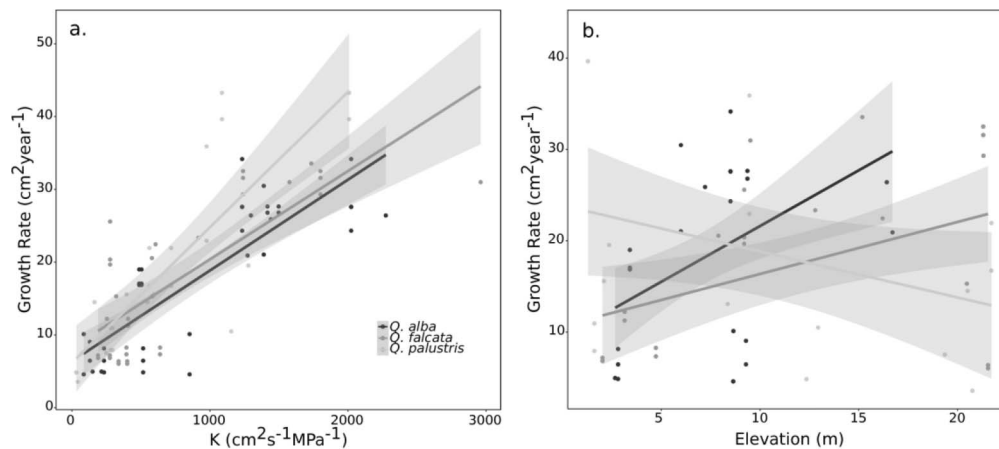


Figure 6. Growth rate (BAI, $\text{cm}^2\text{year}^{-1}$) versus (a) hydraulic conductance (K , $\text{cm}^2\text{s}^{-1}\text{MPa}^{-1}$) and (b) elevation (m) measured in mature trees. There is a strong, positive linear relationship between growth rate and hydraulic conductance (Multiple $R^2 = 0.67$), and the slope of that relationship is significantly higher ($P = 0.0312$) in *Q. palustris*, which also exhibits a significantly different ($P = 0.0178$) relationship between growth rate and elevation than the other two species.

Stress Pre-dawn water potential (Ψ_{PD}) (see Figure S8 available as Supplementary Data at *Tree Physiology* Online) was significantly lower at all elevations and for all mature trees in the dry year, reflecting significant differences in measured soil moisture. Mid-day water potential (Ψ_{MD}) was also significantly different between years. Here we focus on the difference ($\Delta\Psi$) between these values as an indicator of plant stress to incorporate information about both changing water availability and changing evaporative demand. Figure 5b illustrates the changes among species in $\Delta\Psi$ between years and elevation categories in mature trees. All species exhibited higher mean $\Delta\Psi$ in the dry year, though this general trend was not observed at all sites. The differences between years were most pronounced at dry sites, and the difference in $\Delta\Psi$ at wet and dry sites was significant in the dry year ($P = 0.007$, Tukey's HSD) but not the wet year ($P = 0.25$, Tukey's HSD). The compounding effect of the drought and elevation gradient on water stress appeared most pronounced in *Q. palustris*, though differences among species in mature trees were not significant.

Quercus palustris seedlings also exhibited higher levels of $\Delta\Psi$ in the low water treatment than at high water, as did *Q. alba*. Differences in $\Delta\Psi$ between water treatment categories and species were significant (ANOVA, $P = 0.0005$ and $P = 0.04$, respectively). The interaction between species and water treatment was not significant when *Q. alba* was included, but differences among species in which water treatment caused the most stress were significant when only *Q. palustris* and *Q. falcata* seedlings were compared ($P = 0.05$).

Stress response measured by chlorophyll fluorescence (F_v/F_m), however, suggested that seedlings exhibited increased photoinhibition (F_v/F_m below 0.8) at high water treatments (ANOVA, $P < 0.0001$). This response was seen in both *Q. falcata* and *Q. palustris*, but not in *Q. alba* (Figure 7).

Growth and productivity There was a weak ($P = 0.08$, linear least squares regression), positive relationship between increasing growth rate (average BAI per year) and elevation, shown in Figure 6b, suggesting slightly higher growth rates in individuals at drier sites at higher elevation. *Quercus palustris*, however, had a significantly different ($P = 0.02$) relationship than the other two species, having its fastest growth in the floodplain and decreased growth rate at higher elevation.

In seedlings, the relationship between growth and water availability was not as strong; all three species produced less total biomass at both high water and low water availability, as compared with the moderate water treatment (Figure 7). However, the differences in biomass between high and low water were not significant, nor were the differences among species. *Quercus palustris* seedlings did show a different pattern of biomass allocation than the other two species, with significantly lower allocation of biomass below ground. This was especially apparent in the high water treatment. There was a significant difference in carbon assimilation (A) among treatment groups ($P < 0.0001$), with plants given moderate water showing higher A than those at high water ($P < 0.0001$, Tukey's HSD). There were no significant differences in A among species, nor was a significant interaction between species and treatment observed.

Mature trees in all three species did exhibit a strong ($P < 0.0001$), positive relationship between hydraulic conductance (K) and average annual growth rate, shown in Figure 6a. The slope of this relationship was significantly ($P = 0.03$) higher in *Q. palustris* than the other two species, meaning for the same increase in K , *Q. palustris* had a larger annual BAI. In *Q. palustris*, this increased slope was the same at both dry and wet sites (slope = 0.02 at wet site and 0.01 at dry site, $P = 0.23$); by contrast, *Q. falcata* had a steeper slope at dry site (increasing

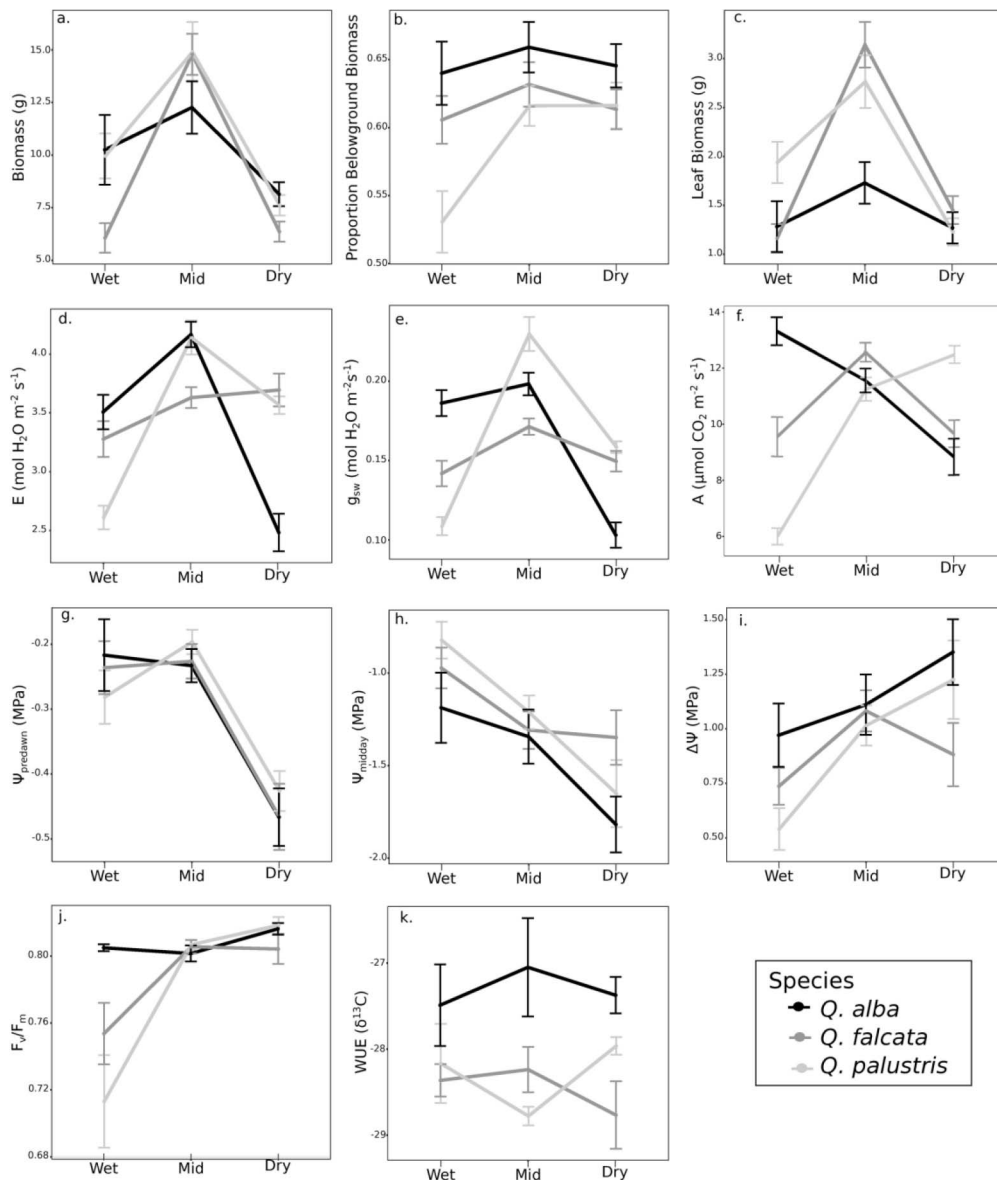


Figure 7. Seedling traits demonstrating (from top to bottom): growth (a, total biomass; b, proportion belowground biomass; and c, leaf biomass), gas exchange (d, transpiration rate; e, stomatal conductance; and f, photosynthesis), stress response and water status (g, Ψ_{PD} ; h, Ψ_{MD} ; i, $\Delta\Psi$; and j, F_v/F_m), and water use (k, $\delta^{13}C$). Significant differences by treatment and species are summarized in Table 5.

from 0 to 0.016, $P = 0.04$) and *Q. alba* had a lower one (decreasing from 0.02 to 0.002, $P = 0.07$), suggesting that each species responded differently to environmental conditions that vary with elevation.

Discussion

These results provide evidence that the three most abundant species of oaks in the 'BTP' at SERC, *Q. alba*, *Q. falcata* and *Q. palustris*, partition an elevation (and thus hydrologic) gradient. *Quercus palustris* and *Q. falcata* showed the largest differences in local distributions, while *Q. alba* was more evenly distributed from low to high elevation. The difference in local distribution

between *Q. falcata* and *Q. palustris* was also reflected in differences in growth rate, with *Q. palustris* experiencing its highest growth rates at low elevation, in the floodplain, where *Q. falcata* exhibited its lowest growth rates. This habitat partitioning was supported in part by climatic differences in the broad geographic ranges of these species, as well as key differences in functional traits of both mature trees and seedlings. Considered in sum, these results underline the complexity of the factors that drive local species distributions in natural systems.

Differences in climatic conditions across the broad geographic ranges of the three species were associated with local habitat partitioning, as hypothesized in Ackerly (2003) and Cavender-Bares et al. (2016), particularly in WI and MAT. The

overall geographic range of *Q. falcata* was associated with significantly warmer and more arid climate than the other two species, despite higher overall rainfall, and locally was found in sites with the lowest seasonal water availability. *Quercus alba* had a broad geographic and climatic range that largely overlaps the other two species, a trend consistent with the local distributions at SERC. However, the distribution of oaks in the BTP showed significant separation between *Q. palustris* and the other two species, while the full climatic distributions of *Q. palustris* and *Q. alba* across their ranges showed no significant differences in average temperature, rainfall or aridity. One caveat to note is that we may not have completely captured the full North American ranges of these species with GBIF data alone. Beck et al. (2014) demonstrated that the spatial bias in specimen records can significantly skew range reconstructions, and the accuracy of species identification and the precision of geographic location can introduce error (Goodwin et al., 2015; Wicczorek et al., 2004). Nevertheless the consistency between aridity and temperature ranges and habitat preference at SERC provides compelling evidence of a connection between broad scale and local distributions.

Among plant functional traits, the hydraulic conductance of mature trees supported the hypothesis that contrasting water-use strategies help *Q. palustris* and *Q. falcata* exploit opposite ends of a hydrologic gradient. With decreasing water availability, in both a dry and wet year, the hydraulic conductance in mature trees of *Q. palustris* declined while that of *Q. falcata* increased or did not change. The conductance of *Q. palustris* also declined consistently with decreasing water in seedlings, while conductance in *Q. falcata* showed a small increase in the moderate water treatment, though the mean values were not significantly different among seedlings of different species. Conductance has been shown to be highly correlated with growth rate, a trend seen in both the literature [e.g., Poorter et al., 2010] and in our data (Figure 6a). In our study, the relationship between growth and conductance was strongest in *Q. palustris*; the same increase in conductive ability was associated with a significantly larger increase in growth rate in *Q. palustris* than the other two species, suggesting it is adapted to take advantage of greater water availability.

The higher conductance seen in *Q. palustris* may come at the cost of increased stress during drought years. Values reported in the literature show that *Q. palustris* typically has a larger vessel diameter than *Q. falcata* (Lobo et al., 2018; Robert et al., 2017), which suggests it may be trading high hydraulic efficiency when water is available for increased vulnerability when it is not (Sperry et al., 2008). We observed mature *Q. palustris* trees experiencing the largest difference between predawn and midday water potential, indicative of drought stress, at the highest elevation in the dry year. $\Delta\Psi$ was also higher in *Q. palustris* seedlings when water was limited, and higher in *Q. falcata* without water limitation.

Other traits in both mature trees and seedlings also provided evidence of partitioning. *Quercus palustris* seedlings allocated less of their total biomass belowground, especially at high water availability, indicating a relatively shallow rooting system compared with the other species. Shallow roots have been shown to significantly decrease the survival of tree seedlings in drought (Padilla & Pugnaire, 2007), and at a global scale lower rooting depth is associated with wet environments (Canadell et al., 1996). Though both species demonstrated an increase in WUE with decreasing water, the change in WUE in drier conditions was the smallest in *Q. falcata*, consistent with the species' preference for drier habitats.

We found limited evidence to support the hypothesis that *Q. alba*'s broader distribution would be supported by an ability to maintain homeostasis under stress. In the high water treatment, both *Q. palustris* and *Q. falcata*, but not *Q. alba*, seedlings exhibited stress (as indicated by low F_v/F_m). Though it might be unexpected to see a water-loving species like *Q. palustris* exhibit higher stress in very well-watered conditions, even wetland species show decline in photosynthetic rates in response to the oxygen deprivation caused by flooding (Pezeshki, 2001), and lowered F_v/F_m is a documented response to flooding stress (Nash & Graves, 1993). In general, oaks show a preference for drier, well-drained soils and do not grow in the wettest climates in their geographic ranges (Cavender-Bares et al., 2018).

The depressed performance of seedlings under high water conditions illustrates that drought was unlikely to be the only environmental factor affecting plant function. Lower VPD or light limitation in the rainy season may have augmented plant response to water stress, helping to explain why we observed a smaller response than expected to the drought conditions of the dry year. We found sap flux to be correlated with both light levels and VPD; it is possible that decreases in transpiration from lower water availability were offset by stronger driving gradients in the dry year, explaining why we did not see a change in TDWL. Aranda et al. (2005) found a higher stress response to drought in cork oak (*Quercus suber* L.) in low light conditions compared with high light. High light could alternatively cause depressed performance or stress due to photoinhibition (Long et al., 1994), which we were not able to test for in this study.

Finally, while three species are too few to draw strong conclusions about the effects of phylogeny on habitat sorting, the distributions of these three oak species at SERC did match the expectation that more closely related species will show greater habitat separation, drawn from microcosm experiments (Violle et al., 2011), across environmental gradients (Cavender-Bares et al., 2004a, Fallon & Cavender-Bares, 2018) and on continental-scale observations of different subgenera (Cavender-Bares et al., 2018). *Quercus palustris* and *Q. falcata*, both red oaks (Section *Lobatae*) are more closely related to each other and more separated along the elevation gradient than either is from *Q. alba*, a white oak (Section *Quercus*). The pattern

suggested by these three oak species was consistent with phylogenetic relatedness as a driver for community structure and functional diversification.

In addition to concerns about the accuracy of GBIF data and limited power for phylogenetic analysis, there are a few cautions that may limit the scope of these results. Heat dissipation sap flow measurements may fail to accurately estimate transpiration rate, in particular because there can be a high amount of radial and circumferential variation in sap flow that may not be captured by one or two measurements per tree (González-Altozano et al., 2008); we observed a high degree of dispersion in results from pairs of sensors installed in the same tree (see Figures S5 and S6 available as Supplementary Data at *Tree Physiology* Online), which could be attributed to natural variation or could be an artifact of differences between sensors (Lu et al., 2004). Bush et al. (2010) have observed that the calibration constants published by Granier (1987) may not be accurate for ring-porous tree species, like oaks. Second, the design of this study was not optimal for testing competition or local adaptation directly. This was not a reciprocal transplant experiment, nor were genotype or maternal line controlled (Bengtsson et al., 1994). Our results should, however, motivate further research that does explicitly control these variables, because the distribution pattern and trait differences we found warrant additional investigation.

Conclusion

We have provided evidence supporting the hypothesis that *Q. alba*, *Q. falcata* and *Q. palustris* coexist in the forest community at SERC by partitioning a hydrologic gradient driven by elevation. Our findings suggest that a combination of biogeographic legacy effects, functional traits, response to temporal variation and phylogeny may play a role in driving this variation. Among functional traits, the hydraulic conductance of mature trees offers the clearest support to the idea that the two red oaks partition the gradient through contrasting water-use strategies. The fact that no single phenomenon among those we tested can explain the local distribution of our study species is consistent with other recent work (Morueta-Holme et al., 2016), and future research in habitat partitioning and community assembly will be strengthened by addressing multiple potential drivers of observed patterns.

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